

CS 498 HW0 Report

Perspective, Homography, and Epipolar Geometry

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Fall 2026

1 Homography and 2D insertion

Normalization. Before building any linear system I precondition each 2D point set separately. For a set with centroid $\mathbf{c}$ and mean distance d from that centroid, the similarity

$$T = \begin{bmatrix} s & 0 & -s c_x \\ 0 & s & -s c_y \\ 0 & 0 & 1 \end{bmatrix}, \quad s = \sqrt{2}/d,$$

moves the centroid to the origin and scales the mean radius to $\sqrt{2}$. I build T_{xy} from the court points and T_{uv} from the pixels. Without this step the columns of the design matrix range from 1 to $u \cdot x \approx 10^4$ and the smallest singular vector is poorly determined.

DLT. With $\mathbf{p} = [x, y, 1]^\top$ and H written as rows $\mathbf{h}_1, \mathbf{h}_2, \mathbf{h}_3$, the relation $\lambda[u, v, 1]^\top = H\mathbf{p}$ loses λ by substituting the third component into the first two, giving two rows per point,

$$\begin{bmatrix} -\mathbf{p}^\top & \mathbf{0}^\top & u \mathbf{p}^\top \\ \mathbf{0}^\top & -\mathbf{p}^\top & v \mathbf{p}^\top \end{bmatrix} \begin{bmatrix} \mathbf{h}_1 \\ \mathbf{h}_2 \\ \mathbf{h}_3 \end{bmatrix} = \mathbf{0}.$$

Stacking all points gives a $2N \times 9$ matrix A . The right singular vector of A with the smallest singular value is the unit vector minimizing $\|A\mathbf{h}\|$; I reshape it row-major to 3×3 , undo the normalization with $H = T_{uv}^{-1} H' T_{xy}$, and divide by the Frobenius norm.

Logo placement. A logo pixel (p_x, p_y) in an $h \times w$ image is first mapped to the metric rectangle with lower-left corner (x_0, y_0) and size (W, L) , then through H :

$$M_{\text{logo} \rightarrow \text{court}} = \begin{bmatrix} W/w & 0 & x_0 \\ 0 & -L/h & y_0 + L \\ 0 & 0 & 1 \end{bmatrix}, \quad M_{\text{logo} \rightarrow \text{image}} = H M_{\text{logo} \rightarrow \text{court}}.$$

The negative entry flips the vertical axis because pixel rows grow downward while court y grows upward. The provided bilinear inverse warp uses the composed matrix. Compositing is the handout equation applied per pixel and channel, $I_{\text{out}} = \alpha I_{\text{logo}} + (1 - \alpha) I_{\text{court}}$, where α is the fourth channel of the warped logo.

Results. The maximum corner-transfer error on the four basketball corners is 5.1×10^{-13} px, which is zero up to floating point. Four points give eight equations for eight unknowns, so the fit is exact by construction.



(a) 1A: given corners as crosses, H -projected corners as circles.



(b) 1B: 6×3 m logo composited at lower-left (23.0, 2.5) m.

Figure 1: Basketball homography fit and logo insertion.

1.1 Tennis transfer and camera height

The course frame puts the origin at the center of the near baseline with $+x$ across the court and $+y$ toward the far baseline. Using the regulation length 23.77 m, doubles width 10.97 m and singles width 8.23 m, the metric coordinates for the supplied anchors are:

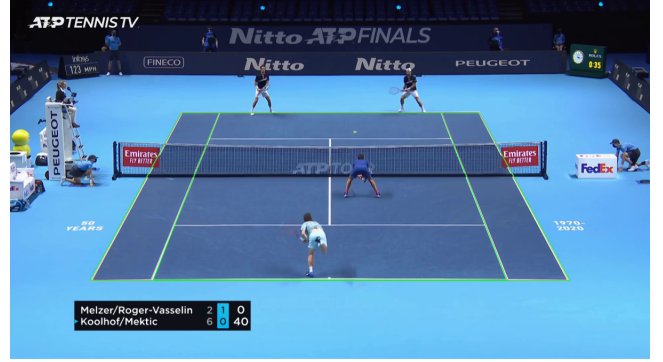
	intersection	x (m)	y (m)
A	near left doubles corner	-5.485	0
B	near right doubles corner	5.485	0
C	far left doubles corner	-5.485	23.77
D	far right doubles corner	5.485	23.77
E	near left singles corner	-4.115	0
F	near right singles corner	4.115	0
G	far left singles corner	-4.115	23.77
H	far right singles corner	4.115	23.77

Running the same three functions on frame 0 of `match005.mp4` gives an eight-anchor reprojection RMSE of 3.67 px (Fig. 2). The near anchors carry 4 to 6 px and the far anchors about 1 px.

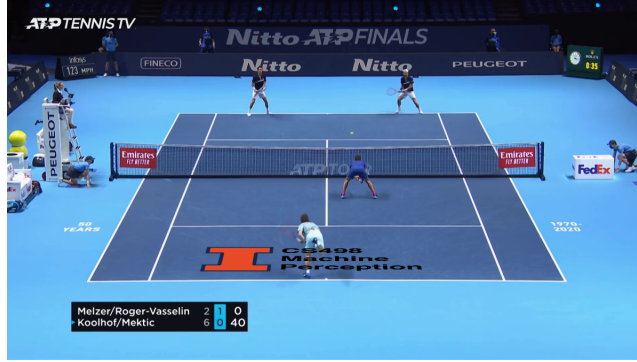
Camera height. For a planar scene the camera matrix loses its third column, so $H \sim K[\mathbf{r}_1 \ \mathbf{r}_2 \ \mathbf{t}]$; $\mathbf{r}_3$ never appears because every court point has $Z = 0$. With K unknown this product cannot be split apart. A floor point projects to $u = f X_c/Z_c + c_x$, so moving the camera twice as far away doubles Z_c and doubling the focal length doubles f ; the two changes cancel and every floor pixel stays put. Two cameras at different heights therefore produce the same H . Counting degrees of freedom says the same thing: H has 8, while K (5), R (3) and $\mathbf{t}$ (3) have 11, so a three-parameter family of cameras shares each H . The basketball scene breaks the tie with the four backboard corners at $Z > 0$. Those points constrain the column of P that the plane discards, so the full 3×4 matrix, and from it K , R and the height, become observable without knowing K in advance.



(a) Supplied anchors A–H.



(b) Given anchors in green, reprojected in orange.



(c) Logo in $[-3, 3] \times [0.5, 3.5]$ m.

Figure 2: Tennis transfer using the coordinates above and unchanged Task 1 code.

2 Camera projection and 3D insertion

Projection DLT. The construction is the Task 1 DLT with a 3D point. With $\mathbf{X} = [X, Y, Z, 1]^\top$ and P written as rows $\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3 \in \mathbb{R}^4$, each landmark contributes the rows $[-\mathbf{X}^\top, \mathbf{0}^\top, u\mathbf{X}^\top]$ and $[\mathbf{0}^\top, -\mathbf{X}^\top, v\mathbf{X}^\top]$ to a $2N \times 12$ system. I normalize the 3D points with a 4×4 similarity that scales the mean radius to $\sqrt{3}$ and the pixels with the 3×3 transform from Task 1, take the last row of $V^\top$, reshape to 3×4 , undo with $P = T_{uv}^{-1}P'T_{xyz}$, and fix the Frobenius norm. P has 11 degrees of freedom so at least 6 points are needed; all eight landmarks are used.

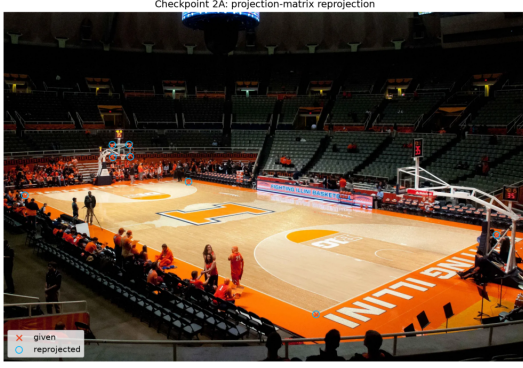
Role of K . If every landmark had $Z = 0$, the four Z columns of the $2N \times 12$ design matrix would be identically zero. The three entries of P in its Z column would then multiply nothing, so any values for them satisfy $\mathbf{A}\mathbf{p} = \mathbf{0}$ and the null space has dimension at least four; the SVD returns an arbitrary vector from it and the problem collapses to the homography of Task 1. Whether that is a problem depends on K . If K is known, the plane is enough: $K^{-1}H = [\mathbf{r}_1 \ \mathbf{r}_2 \ \mathbf{t}]$ up to scale, and $\mathbf{r}_3 = \mathbf{r}_1 \times \mathbf{r}_2$ completes the rotation. If K is unknown, as here, H cannot be factored and points off the plane are needed to pin down all four columns of P . Deleting the Z column of my estimated P gives back my Task 1 H to within 0.0014 after scaling both to unit norm, which confirms the two tasks estimate the same camera.

Results. The mean and maximum landmark reprojection errors are 2.75 px and 5.10 px (Fig. 3). The court corners are within 1.7 px; the four small, distant backboard corners carry 4 to 5 px. The provided

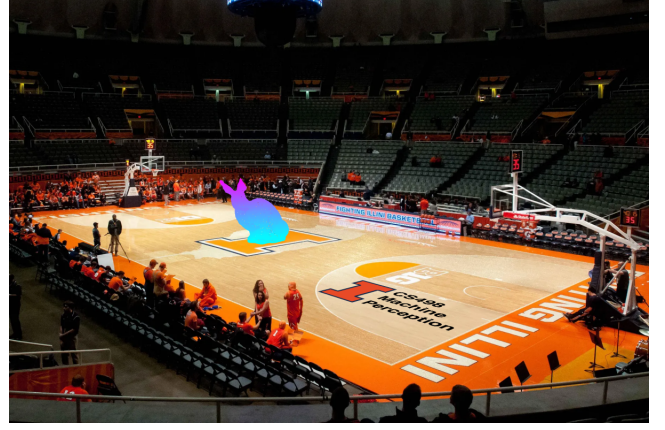
decomposition returns

$$K \approx \begin{bmatrix} -763.9 & -258.9 & 481.8 \\ 0 & -361.8 & -202.2 \\ 0 & 0 & 1 \end{bmatrix}, \quad R \approx \begin{bmatrix} -0.706 & -0.708 & -0.019 \\ 0.602 & -0.614 & 0.510 \\ -0.372 & 0.349 & 0.860 \end{bmatrix}, \quad \mathbf{t} \approx \begin{bmatrix} 22.72 \\ -39.86 \\ 17.21 \end{bmatrix},$$

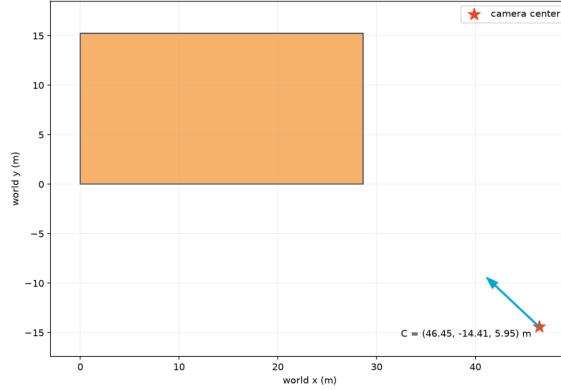
with $C = (46.45, -14.41, 5.95)$ m, so the optical center sits 5.95 m above the court. Loading `task2_results.npz` and computing $-R^\top \mathbf{t}$ reproduces C to 7×10^{-15} m per component; $R^\top R = I$ with $\det R = 1$; and $P[C; 1] = \mathbf{0}$, so C is the null vector of P . Following the handout, K , R and $\mathbf{t}$ are treated as noisy DLT estimates rather than a calibration; the negative focal terms and large skew in K reflect that.



(a) 2A: eight landmarks reprojected by P .



(b) 2B: bunny at center court, colored by height.



(c) 2C: recovered camera center and viewing direction.

Figure 3: Camera matrix estimation, 3D insertion, and pose.

3 Two-view epipolar geometry

Method. Each match row is (u_1, v_1, u_2, v_2) . The two views are normalized independently with T_1 and T_2 as in Task 1. The epipolar constraint $\mathbf{x}_2^\top F \mathbf{x}_1 = 0$ is already linear in F , so each match gives a single row $[u_2 u_1, u_2 v_1, u_2, v_2 u_1, v_2 v_1, v_2, u_1, v_1, 1]$ of an $N \times 9$ matrix. The last row of $V^\top$ reshaped to 3×3 is $\tilde{F}$. A second SVD enforces rank two, $F' = U \text{diag}(\sigma_1, \sigma_2, 0) V^\top$, which is the nearest rank-2 matrix in Frobenius norm. Because the constraint is bilinear the denormalization is $F = T_2^\top F' T_1$, a transpose rather than an inverse. F is then scaled to unit Frobenius norm.

Results. From the eight supplied matches,

$$F \approx \begin{bmatrix} -2.20 \times 10^{-9} & 1.06 \times 10^{-7} & -1.38 \times 10^{-4} \\ 2.68 \times 10^{-7} & 2.54 \times 10^{-8} & 7.05 \times 10^{-4} \\ -2.97 \times 10^{-4} & -1.46 \times 10^{-3} & 1.00 \end{bmatrix}, \quad \text{rank } F = 2.$$

Over all 200 evaluation matches the symmetric epipolar distance has mean 5.58 px and median 4.52 px (Fig. 4).

Why the residuals are nonzero. Four effects keep the residuals away from zero. The 200 keypoints come from a feature detector on real photographs and are localized to roughly a pixel, so each match carries noise before F is estimated. F was fitted to only 8 of the 200 matches, and the other 192 cannot be predicted better than the fit itself. The rank-2 projection moves F off the exact null vector, so even the 8 fitted points no longer lie exactly on their lines; their mean error is 1.7 px. Finally, F assumes ideal pinhole cameras and cannot absorb lens distortion. Refitting with all 200 matches lowers the mean to 2.8 px, which shows the remaining error is a property of the data rather than the code.

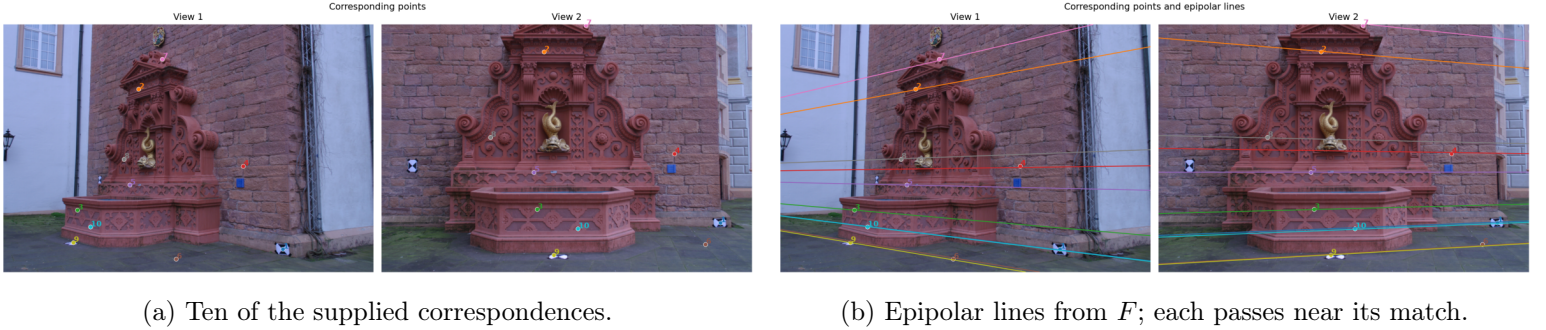


Figure 4: Two-view geometry on the fountain pair.

4 Limitations and improvements

Annotation precision. The basketball error is exactly zero only because four points give eight equations for eight unknowns; any click error is absorbed into H instead of appearing as residual, so that number says nothing about accuracy. The tennis (3.7 px), landmark (2.8 px) and epipolar (5.6 px) fits are overdetermined, and their residuals are the click and detection noise finally becoming visible. The near-baseline tennis anchors are the worst because perspective spreads a pixel of error there over more meters of court.

Degenerate configurations. The homography fails if three or more of the points are collinear. The projection DLT fails when all landmarks are coplanar, the $Z = 0$ case discussed in Section 2. The eight-point algorithm fails for a pure rotation between views, since without a baseline there are no epipolar lines, and when all matched points lie on one plane, since a homography then explains the matches and F is not unique.

Improvements. First, the DLT minimizes an algebraic residual rather than pixel distance; a few iterations of nonlinear least squares on reprojection or Sampson error after the linear solve would reduce the geometric error directly. Second, F should be estimated from all 200 matches with RANSAC rather than from eight hand-picked ones, so that a large inlier set averages out the detection noise. Third, a radial distortion model for the broadcast tennis frames would remove the systematic near-versus-far error pattern.

Optional foreground-preserving video bonus

I implemented the complete bonus pipeline in `bonus_tennis_video.py` without any provided helper. Video loading, background estimation, the foreground mask, logo insertion, composition and output writing are all in that file; the only reused code is my Task 1 `estimate_homography`, `logo_to_image_homography` and `alpha_blend`.

Background estimate. Every clip has a fixed camera, so the per-pixel temporal median of 45 frames sampled evenly across the clip removes the moving players and leaves the empty court.

Foreground alpha. For each frame I take the maximum absolute color difference to that background, pass it through a smooth ramp between 0.10 and 0.28, clean it with a morphological opening (speckle) and closing (holes inside players), dilate slightly, and Gaussian blur to feather the edges. The result is a soft mask $m \in [0, 1]$.

Composition. The logo is warped with the Task 1 homography built from the clip’s eight supplied image anchors and the metric A–H coordinates above, dimmed by the ratio of live to background luminance so shadows fall across it, and blended with effective alpha $\alpha_{\text{logo}}(1 - m)$. Where a player is present the effective alpha drops to zero and the original frame pixel remains on top; the player is never cut out and pasted back. Compositing runs only inside a box around the logo footprint. The representative frame for each clip is the one where m overlaps the logo most (Fig. 5), and `match005_o_cclusion_aware.mp4` holds all 250 composited frames.

AI-use declaration

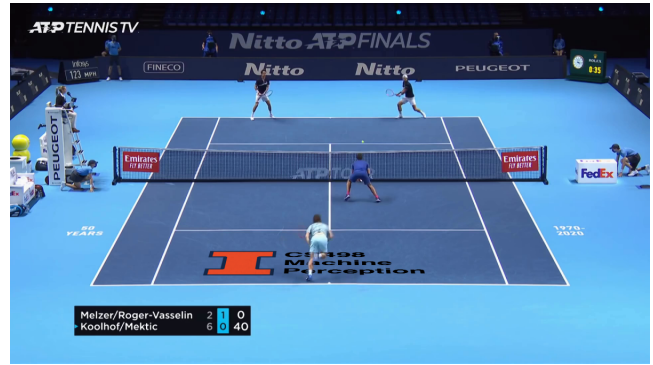
Check one statement.

- ☐ I did not use generative AI for this homework.
- ☒ I did not use generative AI for any required part. I used it only in the optional bonus marked “Gen-AI leverage allowed.” Tool/model, purpose, and affected files: Claude (Anthropic), used to design and write the bonus compositor; affected files are `bonus_tennis_video.py` and the generated `outputs/bonus/` only.

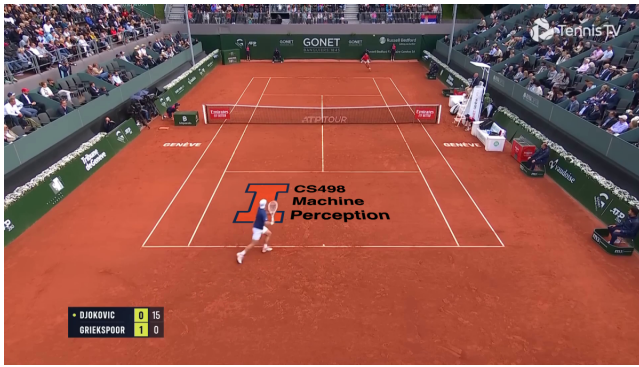
I confirm that generative AI did not write or complete my required code, select required correspondences, derive my required answers, generate my analysis, or write my required report.



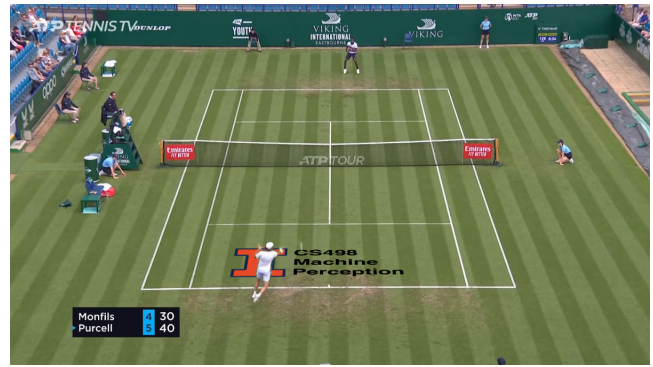
(a) match001, frame 500.



(b) match005, frame 3.



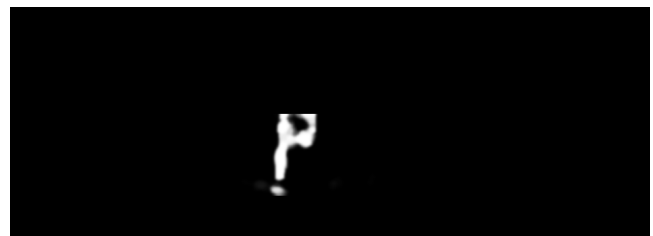
(c) match028, frame 131.



(d) match091, frame 356.



(e) match125, frame 578.



(f) Soft mask for the match005 frame, logo region.

Figure 5: Occlusion-aware insertion: the player stays in front of the logo in every clip.